

TRAINING GUIDE

*TRAINING
THAT
WORKS*



Using Education as a Catalyst for Change

How Noria Transformed Its Training Model to Revolutionize the Industry
Learn More on Page 4. >>



WHY CHOOSE NORIA?

The Difference is in the Details

It's been said that the only constant thing is change. At Noria, we realize change is vital for a company to thrive and sustain for the future. Positive change is impacted through a variety of means, but one important vehicle for change is education.

We are the Source.

Noria is the global authority in machinery lubrication and oil analysis. Each year, we train thousands of professionals worldwide – and in more than 10 different languages – on innovative plant reliability measures through excellence in machine lubrication. Our training encompasses all types of mechanized industries.

We Share Your Passion.

Noria continues to pursue a mission of inspiring change through education by providing customers with the tools, resources and expertise to achieve the tremendous, untapped potential that lubrication excellence has to offer.

We Have Been Down This Road Before.

Noria's team of experts combines a wide variety of industry experience in reliability, engineering and lubrication excellence to provide world-class training and expertise. Whether you need industry education, technical consulting, failure investigation or program/analysis development, our experts can help you.

Along with extensive experience, Noria's team relies on a large network of industry leaders to stay current with industry technologies, trends and client needs. Our team works hard to bring the best services and highest level of expertise to each client.



We Work Together with You.

Time is money, and your time is valuable. To accommodate our customers, Noria offers a variety of training options that includes online and private training. We can also customize course curriculum and training to achieve your desired results.

We Implement Change that Endures.

The difference is in the details. Noria's courses are designed to align with the needs of adult learners, as determined by expert staff that includes an adult-learning specialist, an instructional designer and multiple graphic artists and video/multimedia developers. Training sessions are goal-oriented, relevant and practical. Interactive opportunities are incorporated when possible to maximize knowledge retention and understanding.

We Want You to Succeed.

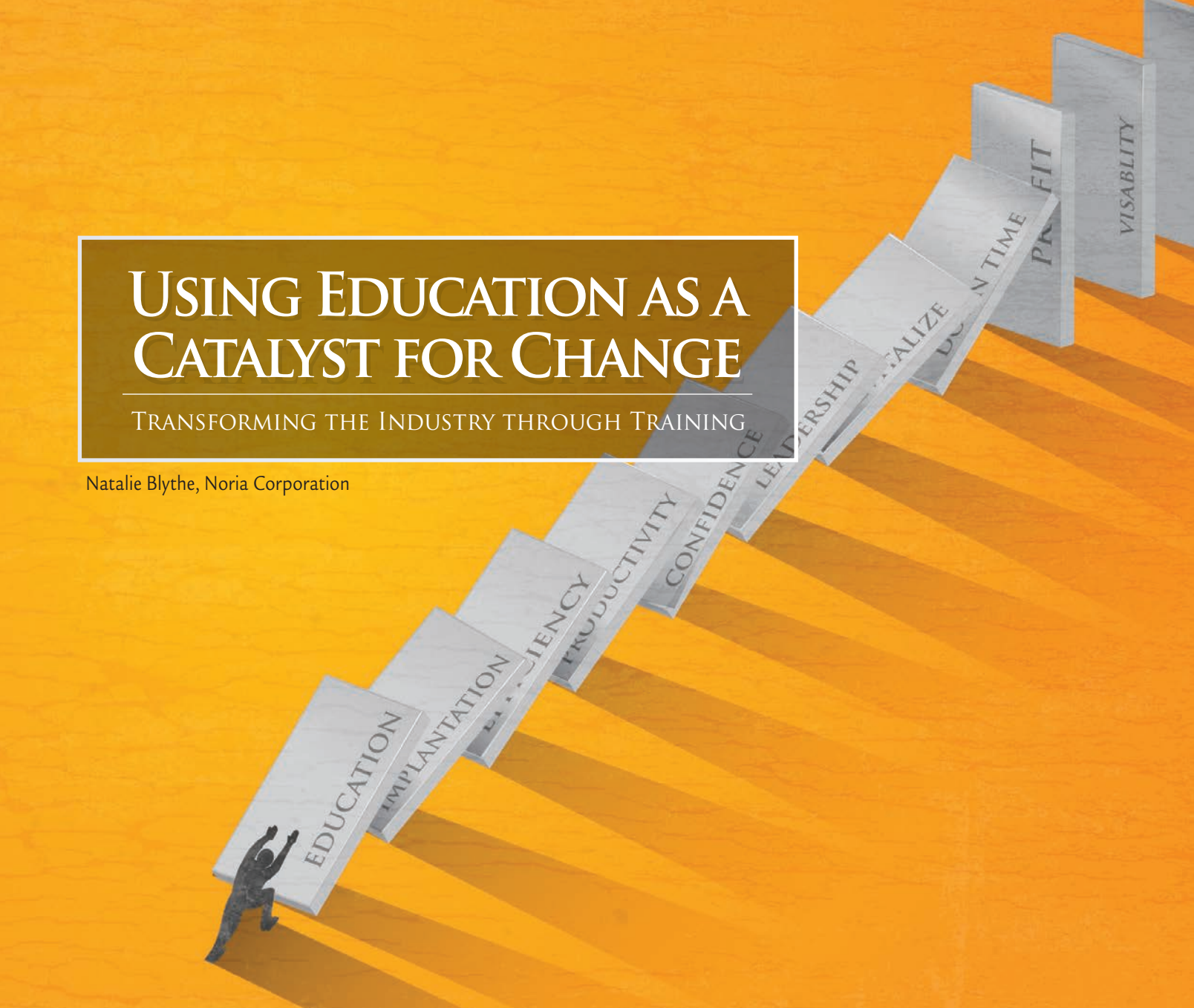
All course material – from beginner to advanced – is task-based and immediately applicable in the workplace. Students receive preventive maintenance tips and learn why every strategy is important. Knowledge is powerful and when our workforce becomes empowered with excellent lubrication methods, the results are a safer, more efficient and more reliable industry.

We Guarantee Satisfaction.

Noria proudly stands behind our training courses. If you're not satisfied with the information and insights you gain at these events, we will refund 100% of your paid registration fee.

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USING EDUCATION AS A CATALYST FOR CHANGE

TRANSFORMING THE INDUSTRY THROUGH TRAINING

Natalie Blythe, Noria Corporation

“The only thing that is constant is change,” said Heraclitus of Ephesus.

This statement has never been truer than today.

According to a recent study by the Manufacturing Institute, more than 75% of manufacturers experienced a moderate or serious shortage of qualified applicants for skilled and highly skilled production positions. The study also found that this skills shortage costs manufacturers up to 11% of net earnings. To combat this, the Manufacturing Institute reported that manufacturers invest approximately \$1,000 in training per employee each year (“Out of Inventory” 2-10).

But what drove this change in education?

“The change came from requests by companies with a reliability focus and industry professionals who needed to get their teams to correctly apply lubrication in the facility. At the corporate level, organizations need training to impact the business, but the question comes down to whether an organization is

“THE ONLY
THING THAT IS
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CHANGE

looking for knowledgeable employees or competent employees,” says Chris Christenson, director of education at Noria.

“The focus for each organization varies with the different roles within the facility, but it can be hard to pinpoint because the requirements are driven by each organization,” he says.

“Some companies do want technicians to have certification as a prerequisite for a job. At this level, there’s already been a pre-awareness for the knowledge, and that education is the impetus for change.”

Christenson has a lengthy history as an educator and instructional course designer. Prior to joining Noria, he was the senior learning manager at the University of Phoenix, leading training facilitation, deployment of operational programs and diagnosis and coaching of staff and leadership for three territories. Before that, he served as the campus college chair for the University of Phoenix and computer graphics chair for the Career Education Corporation.

Having been at Noria for five years, Christenson has thrown himself into revamping the training program. “For the first two years, my time was spent getting our content into a logical flow and working with our instructors on standardizing their approach,” he says.

Prior to Christenson’s arrival, Noria’s training courses needed the addition of interactive experiences which lead to applied learning. This approach greatly increases comprehension and retention. With this focus on the student’s experience, Noria’s courses are aligned with the needs of adult learners – a concept which is gaining momentum within the ever-evolving workplace.

With adult learners, their learning styles are different than traditional students, and the training approach and delivery needs to reflect that. Adult learners, defined as a person who is older than 25 and involved in forms of learning, are typically employed full-time, looking to enhance their professional lives through education and use their personal experience as a resource (Pappas).

Keeping adult learners in mind, Noria divided its training model into three

facets. First, the trainings are designed for utilizing the student’s experience to unlock new knowledge. Then, the trainings help the students internalize the new concepts by relating it to their individual experiences. Lastly, students gain confirmation of the training’s importance through in-class discussions.

“It’s important to have a time where participants reflect, talk about their experiences and have the instructor record those experiences on a board. That way, every time someone references his/her experience, the class can go back to that issue and discuss it,” Christenson says. “It’s a slight position change from telling someone what is good to showing them what is good, and being able to show change is powerful.”

Noria’s model supports evidence from a recent study that showed how adult learners need to make the connection from the curriculum and program content to workplace performance and career development to promote continuous learning behaviors (Yoo and Huang, 2013).

“By using the adult learning approach, we can break down the mindset of ‘We do it this way because this is the way we’ve always done it’ and show how education is catalyst for change,” Christenson says.

With its continual focus on evolving and improving their courses, Noria is also prepared to aid facilities experiencing generational changes in its workforce, particularly since the maintenance and reliability industry is on the cusp of its own “Great Crew Change,” a term recently used during the 2016 downturn in the oil and gas industry that referred to the large workforce generation gap and subsequent talent shortage.

“In five to 10 years, a lot of institutional knowledge – from veteran employees – will be leaving as these

“ **THROUGH
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veterans transition into the next phase of their career, and possibly retirement,” Christenson says. In response, he believes companies are starting to realize the need to change their reliability approach from reactive to proactive. “When bringing in a new workforce, you want to ensure they are learning good, structured processes rather than what passed through the grapevine.”

Christenson’s ultimate goal is for Noria to be the world-renowned source for applied knowledge. “Through knowledge, we want to spark that need for change and with our hands-on courses, we want to show you how to make that change.”



FIND YOUR PERFECT FIT

Something For Everyone

Noria's course plan is versatile. From beginners to experienced technicians and engineers, we can help you no matter what stage you are currently. Follow the entire course track, progressing from beginner to the expert level, or pick and choose according to your specific needs and interests.

Start here if you have working knowledge or moderate experience with machine maintenance

Start here if you have little or no experience with machine lubrication.

Industrial
Lubrication
Fundamentals

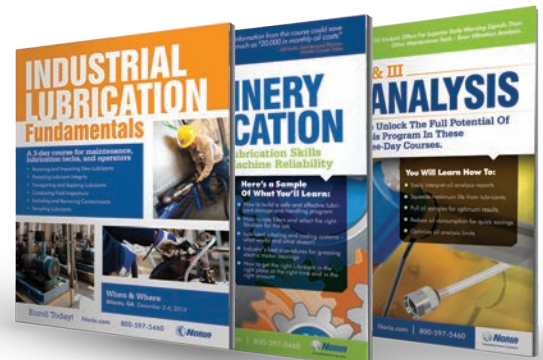
Machinery
Lubrication **I**

Oil
Analysis **II**

Machinery
Lubrication **II**

Oil
Analysis **III**

The course progression shown is recommended but not required. To determine the best course for you, contact us at **918-392-5063** or email training@noria.com.



Basic to Most Advanced

Noria's courses are built from the same foundational concepts, layering more detailed explorations or specialized topics in higher level coursework. The chart below gives a glimpse into which topics are emphasized in each course as well as how technically detailed the information is. **Topic areas are classified from the most basic (◆) to the most advanced (◆◆◆◆).**

Topic Area	Industrial Lubrication Fundamentals	Machinery Lubrication I	Machinery Lubrication II	Oil Analysis II	Oil Analysis III
Maintenance Strategies	◆	◆◆◆◆	◆	◆	◆
Lubrication Concepts	◆◆	◆◆◆◆	◆	◆	◆◆◆◆
Fundamentals of Machine Wear	◆	◆◆	◆◆	◆	◆◆◆◆
Lubricant Selection	◆	◆◆	◆◆◆◆	◆	◆
Reception and Storage	◆◆◆	◆◆	◆◆◆◆	N/A	N/A
Handling and Application	◆◆◆◆	◆◆◆	◆◆◆	◆	◆
Contamination Control	◆◆	◆◆	◆◆◆◆	◆◆◆◆	◆
Lubricant Analysis	◆	◆	◆◆	◆◆◆◆	◆◆◆◆
Wear Debris Analysis	N/A	N/A	N/A	◆◆◆◆	◆◆◆◆
Oil Analysis Program	N/A	N/A	N/A	◆◆	◆◆◆◆
Environmental Disposal	◆◆◆◆	◆◆	◆	N/A	N/A
Certification Preparation	N/A	MLT I and MLA I	MLT II	MLA II	MLA III
Availability	Private Classroom	Public, Private Classroom & Online	Public, Private Classroom	Public, Private Classroom	Public, Private Classroom

Topic Area Scale From 1-4 ◆



Bridging The Gap

INDUSTRIAL LUBRICATION FUNDAMENTALS

Training Options

- ONSITE

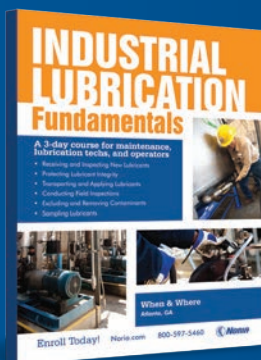
Lay the Groundwork for Future Contributions

Every job has a specialized skillset rooted in its foundation. Industrial Lubrication Fundamentals is the starting point for individuals with all levels of mechanical or maintenance experience. This essential course provides strong techniques and perspective for those with some lubrication knowledge, while presenting sound practical material and instructions for those starting their journey in the field.

Based on extensive collaboration between Noria and plant reliability professionals, the skillset for basic lubrication excellence provides students with the most useful, up-to-date information. This course presents materials in a task-based context, so students receive a well-rounded experience by engaging in interactive exercises, discussing case studies and gaining confidence through hands-on examples.

Who Should Attend?

As an entry-level course, Industrial Lubrication Fundamentals is designed for individuals with little to no technical background, who are responsible for lubrication tasks. Attendee job titles may include lubrication technicians, maintenance technicians, millwrights, mechanics, oilers, machine operators, multicraft technicians, etc.



Want to learn more about this course? Download our brochure or visit **Noria.com/train**.

Course Topics

- Receiving and Inspecting New Lubricants
- Maintaining Lubricant Containers, Machines and Lubricant Storage Areas
- Maintaining Lube Room Inventory
- Removing Contaminants from New Oils with Filter Carts or Installed Systems
- Protecting Lubricant Integrity through Contamination Control
- Transporting and Applying Lubricants
- Conducting Field Inspections on Machines
- Excluding and Removing Contaminants from Lubricated Machines
- Sampling Lubricants
- Importance of Safety, Leak and Spill Response, and Disposal of Hazardous Materials
- Modifying Machinery for Contamination Control, Safety and Reliability Improvement
- Understanding the Communication and Administrative Tasks Necessary for Your Role



Take This Course If You Are Responsible For:

- Electric Motors
- Compressors
- Diesel Engines
- Final Drives
- Gas/Steam Turbines
- Gearboxes
- Hydraulic Systems
- Hydrostatic Transmissions
- Paper Machines
- Process Pumps
- Rolling Mills
- Blowers/Fans

Training That Benefits Your Entire Facility

Training is an ongoing process; not just a one-time event. We realize the need for constant training and made this course easy to take back with you. Upon taking this course, you will receive 90-day access to Noria's video-based procedures used in the course. Each video procedure comes with a printable job aid to ensure you can take the training with you as you head out to the plant floor. With access to our world-class procedures, you can refresh and review the content learned in the course anytime and anywhere.

Topics Include:

- Conducting inspections
- Oil changes
- Electric motor bearing regreasing
- Installation of external oil level indicators, valves, quick connectors and more
- Oil sampling
- Using top-up containers
- Oil filtration using a dedicated filter cart

“Noria produced **instantly usable knowledge**, which will definitely result in **changes** in the way we handle **lubricants and lubrication systems**.”

– Joe Kelly, Maintenance Engineer, Akzo Nobel



1 DAY
TRAINING

NEW
COURSE

FOOD PROCESSING EQUIPMENT LUBRICATION

Best Practices for Safety and Compliance

Course Description

When it comes to food processing, effective lubrication is fundamental to reliable and efficient manufacturing. When the Food Safety Modernization Act (FSMA) was enacted 2011, manufacturers contributing to the creation of food, pharmaceuticals and dietary supplements were mandated to implement systems and controls that specifically address the hazards which impact the safety of the food supply. This new law puts the burden on the individual company's CEO and board of directors to ensure that the food produced is safe. The implementation of the Hazard Analysis and Critical Control Points (HACCP) and the Hazard Analysis and Risk-Based Preventive Controls (HARPC) emerged as primary guiding principles, shifting the focus on responding to contamination in the U.S. food supply to prevention.

Where Does Your Lubrication Program Fit In?

When it comes to the lubrication and lubricant contributions to FSMA, facilities need to implement, follow and maintain several procedures to ensure compliance. These steps range from creating a dedicated maintenance team and documented procedures to using the right lubricants and properly applying them. Facilities also need to understand the various contamination types (biological, physical and chemical threats), along with ensuring their lubrication program uses a lubricant identification system and proper lubricant storage and handling. At the end of the day, you need to ensure that your lubrication program can be held accountable for meeting the FSMA legislation.

Training Options

- Public
- Onsite

Course Topics

- How to develop a plan for assessing current food-grade lubricant use
- How to conduct a Hazard Analysis and Critical Control Points (HACCP) review of your lubrication program
- How to develop compliant standard operating procedures
- How the various food-grade lubricant requirements and consequences of non-compliance contribute to the biological, chemical and physical risks outlined in FSMA
- How to prepare for an FDA inspection and know what you will need for an FDA inspection



Gauge Your Confidence Level

When determining your confidence level, ask yourself these questions:

- How confident are you that your maintenance plan will meet all the requirements of HARPC?
- Can you produce that plan to FDA inspectors?
- Who is your qualified FSMA team?
- Do you have the right lubricants – are they food grade and the right type?
- Eliminate all the guesswork with this course

History Behind the Legislation

After the terrorist attacks on Sept. 11, 2001, the United States government created the Public Health Security and Bioterrorism Preparedness and Response Act (2002) to establish procedures for protecting the public from a threatened or actual bioterrorist attack on the U.S. food supply and other food-related emergencies. Since then, there have been dramatic changes in the global food system, including a more high-tech and complex food supply, shifting demographics and increased need for understanding food-borne illnesses, their consequences and ways to prevent them. The recent enactment of the Food Safety Modernization Act has increased the responsibility of the manufactures to ensure food purity.





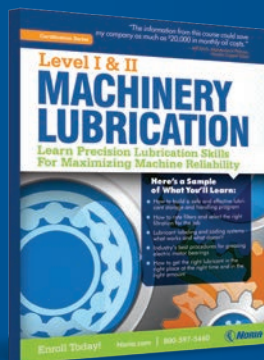
Technical Knowledge Starts with a Solid Foundation

MACHINERY LUBRICATION I

Learn the Crucial Elements of Maintenance, Lubricant Selection and Filtration

Early on, Noria's experts and instructional designers recognized an industry-wide need to establish an effective blueprint for a best-practice-based lubrication program. Machinery Lubrication I (ML I) provides the essentials of proper lubricant application while instilling proven industry methods for selecting, storing, filtering and testing lubricants for improved machine reliability. Through ML I, you will move beyond the dated methods of vague, non-specific lubrication procedures to understand what to do and why there is a right way to do it.

By completing the course, you will leave ML I with a solid understanding of the crucial relationship between lubricant health and machine reliability, as well as understand how doing simple tasks or inspections right will significantly extend machine life and cut costs. You will also gain better understanding of oil analysis, allowing you to align your efforts with those of maintenance professionals or oil analysis experts.



Want to learn more about this course? Download our brochure or visit **Noria.com/train**.

Training Options

- Public
- Online
- Onsite

Course Topics

- How Lubrication Affects Machine Reliability
- Lubrication Fundamentals
- Additives, Base Oils and Grease Thickeners
- Lubricant Performance Properties
- Additive Functions
- Food-grade and Environmentally friendly Lubricants
- Grease and Oil Lubrication Methods
- Lubricants and Fluids for Journal/Rolling-Element Bearings, Gears, Automotive, Compressors, Steam/Gas Turbines, Hydraulics
- Contamination Control
- Oil Drains, Flushing and Reservoir Management
- Storing/Handling Lubricants

Get Certified

Machinery Lubrication I prepares students for ICML Level I MLT certification. Certification testing is available the Friday following the training. Register for your exam at **LubeCouncil.org**.

“Through our Machinery Lubrication training, IKEC initiated a lubrication program that saves them \$2 million each year in annual losses.”

noria.com/serv/customers



Take This Course If You Are Responsible For:

- Lubrication procurement
- Equipment maintenance
- Ensuring equipment readiness and uptime
- Efficiency initiatives/design
- Procedure/quality-control implementation
- Identifying cost-saving methods
- Operations planning
- Equipment reliability

Don't Forget Your Study Packet

Designed for both ICML Level I MLT and Level I MLA certification, the Level I MLT/MLA Study Packet includes:

Flash Card Pack - 385 flash cards designed in preparation for ICML Level I MLT certification.

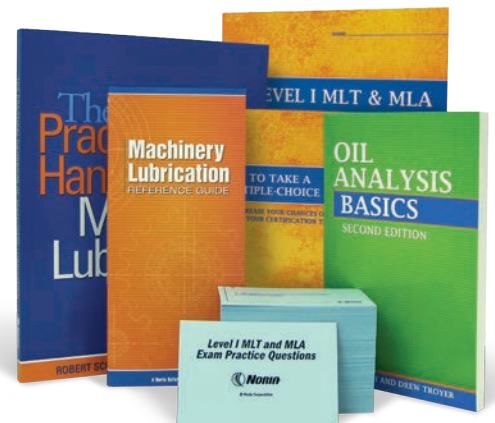
125-Question Practice Exam - A great assessment tool, this multiple-choice practice test is preparation for ICML Level I MLT certification. It is licensed for use by one person.

How to Take a Multiple-Choice Exam - This booklet contains advice from professionals who have passed ICML certification exams as well as helpful hints to prepare the night before the exam, steps to take before entering the exam room; techniques for time management during the exam, and advice for handling diverse types of questions.

Machinery Lubrication Reference Guide - Packed with useful checklists, look-up tables, charts and illustrations, this reference guide is designed to make information easily accessible where you need it. Also doubles as a handy, on-the-job reference tool!

Oil Analysis Basics - With 90+ illustrations, figures and look-up tables, this book presents the fundamentals of oil analysis for machinery condition monitoring in an easy-to-understand format. The book covers everything from how to take a proper oil sample to how to select a test slate for your applications.

The Practical Handbook of Machinery Lubrication - Noria's all-time bestseller, this book contains useful, actionable information that is ready for immediate use. The book contains understandable explanations of how lubricants work, what they're made of and how they break down. Covering topics ranging from engine lubricants to industrial oils and hydraulic fluids, The Practical Handbook of Machinery Lubrication is page-turner that will stay on the shelf.



Add the Study Packet to your course registration, or visit us at store.noria.com.



Advanced Training in Lubrication Excellence

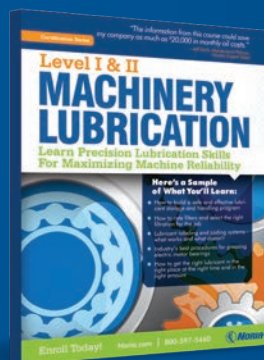
MACHINERY LUBRICATION II

Take Charge of Your Lubrication Program

Machinery Lubrication II (ML II) covers advanced lubrication topics like lubricant selection, troubleshooting, predictive maintenance and more. Designed for maintenance professionals with a solid understanding of the benefits and potential in excellent lubrication practices, this course builds upon their knowledge and provides an in-depth look into the world of lubrication.

Built as a continuation of ML I, the ML II course prompts you to analyze machines for wear symptoms and select lubricants and additives to counteract or prevent damage.

Through ML II, you will be able to distinguish the appropriate lubricants and additives for every application in your facility, guide staff in conducting basic maintenance and analyze machine/fluid indicators for potential problems. These tools will enable you to make comprehensive improvements in the workplace, resulting in key savings of time, money, storage space and training resources.



Want to learn more about this course? Download our brochure or visit **Noria.com/train**.

Training Options

- Public
- Online
- Onsite

Course Topics

- Lubrication Preventive Maintenance Optimization and Design
- Troubleshooting Lubrication Problems
- Lubrication and Oil Analysis Metrics
- Oil Drains, Flushing and Reservoir Management
- Accessorizing New Equipment for Lubrication Excellence
- Lubricating Grease and Oil Application
- Advanced Lubrication Techniques
- Base Oils
- Viscosity and Viscosity Index
- Oxidation and Thermal Stability
- Air Release and Foam Control
- Lubricant Degradation
- Lubricant Selection and Consolidation

Get Certified

Machinery Lubrication II prepares students for ICML Level II MLT certification. Certification testing is available the Friday following the training. Register for your exam at **LubeCouncil.org**.



Take This Course If You Are Responsible For:

- Lubrication procurement
- Equipment maintenance
- Systems operation/management
- Checking meters/recording instruments
- Mechanical inspections
- Controlling lubricant contamination
- Troubleshooting equipment design/process/material
- Equipment reliability
- Production management
- Procedure/quality-control implementation
- Identifying cost-saving methods



For more information
or to register for a
certification exam, visit
LubeCouncil.org.

Get Certified

Certification testing is the Friday following your training and conducted by the International Council for Machinery Lubrication.

Which Certifications?

Completion of Machinery Lubrication I and Machinery Lubrication II will prepare you for Level I Machine Lubrication Technician (MLT) and Level II Machine Lubrication Technician (MLT) ICML certification exams.

What Is ICML?

The International Council for Machinery Lubrication (ICML) is a vendor-neutral, not-for-profit organization founded to facilitate the growth and development of machine lubrication as a technical field of endeavor. In addition to other activities, ICML offers skill certification testing for individuals in the fields of machine condition monitoring, lubrication and oil analysis.

“...the information I learned can
improve our hydraulic systems by **40%.**”

-Vernon Player, PDM Tech, International Paper



2 DAY
TRAINING

NEW
COURSE

OIL ANALYSIS REPORT INTERPRETATION WORKSHOP

Course Description

High-quality oil analysis is one of the most valuable tools in the condition-monitoring toolbox. Oil analysis labs work hard to place critical machine condition data at your fingertips, but it's up to you to take it the rest of the way. In this course, you will learn a systematic approach to translating oil analysis reports into actionable maintenance decisions. You will learn how to select the right tests for measuring the right data points for specific processes, environments and machine conditions. The course also touches on achieving accurate and consistent data collection, removing data noise to accurately identify false results, elemental analysis interpretation, interpreting metallurgical composition combinations to identify wear modes, setting key performance indicators (KPIs) and alarm limits.

Stop Wasting Money on Oil Analysis

Oil analysis isn't cheap, and neither is the equipment on which it reveals information. Every year, industrial plants pay millions of dollars for commercial laboratories to perform analysis on used and new oil samples. Unfortunately, most of the plant personnel who receive these lab reports do not understand the basics of how to interpret them. In this course, you will learn how to mine your oil analysis data to make confident maintenance decisions.

Training Options

- Public
- Onsite

Course Topics

- Using trend report results to identify potential lubricant failures
- Identifying machine wear and failure modes from report data
- Determining when to change, clean or restore lubricant properties
- Identifying when additives have been depleted
- Using elemental analysis results to determine wear locations and mechanisms
- Selecting correct test packages for collecting meaningful data on in-service lubricants
- Extracting meaningful analysis from test reports
- Determining lubricant condition and actions to ensure optimum lubricant performance
- Identifying when the environment and operation have impacted the lubricant
- Applying a systematic approach to reviewing test results and setting next steps
- Identifying data noise from improper lubricant sampling and diagnosing potential sources
- Identifying a quality lab based upon key standards
- Integrating field test and lab analysis results for developing a lubricant health scorecard
- Applying a systematic approach for quick and effective data interpretation
- Interpreting results for determining base oil and additive health, process and environmental contamination, filter performance and machine wear



Are You Catching Critical Early Warning Signals?

Properly established alarm limits are warning flairs that signal your machine is in trouble. There's no time to waste – immediate troubleshooting must take place and the problem found before it is too late. If alarm limits aren't set precisely, the results could be disastrous. Learn how to correctly position alarm limits for specific machine types to ensure you'll receive critical early warnings of impending failures and root cause conditions



Who Should Attend?

- Machinery Lubricant Analysts
- Lubrication Program Managers
- Lubricant Lab Personnel
- Reliability Engineers
- Predictive Maintenance Specialists

If You Use Any Of These Machines

- Gearboxes
- Hydraulic Systems
- Compressors
- Final Drives
- Rolling Mills
- Electric Motors
- Paper Machines
- Diesel Engines
- Blowers/Fans
- Process Pumps
- Gas Turbines
- Steam Turbines

Reduce Data Interpretation Errors

Even if you do everything else right, the success of an oil analysis program always comes down to correct interpretation of test data. In this course you'll learn a systematic approach to translating your oil analysis data into actionable information. You'll learn how to get the most from your oil analysis program and stop jumping to conclusions too early in the process, reducing errors in interpretation.





Developing Verified Data

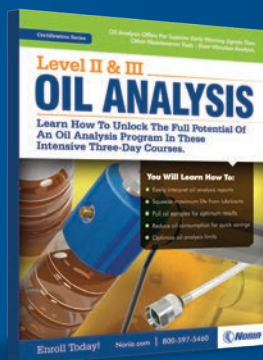
OIL ANALYSIS II

Uncover the Finer Aspects of Contamination Control and Comprehensive Oil Analysis

Oil Analysis II (OA II) approaches lubrication excellence through the critical information to be gleaned from oil analysis. By starting this course, you will understand the basic analysis procedures from ML I, and learn how to combine and apply these for more advanced strategies in conducting impactful predictive maintenance.

In OA II, you will gain everything needed to properly sample, monitor health and detect additive, thermal and particle factors that degrade lubricants via on-site tests and lab analysis. Taking knowledge from an oil analysis lab report, you will develop a proactive line of defense against failures, factoring lab findings into your own onsite analyses for more precise, timely maintenance.

By reviewing case studies, reading and interpreting oil analysis reports and participating in group problem-solving exercises, you will delve into the best practices of oil sampling, learn the fundamentals for conducting effective fluid analysis and become adept in analyzing lubrication systems for optimal efficiency and savings, ensuring that machine/fluid health, lubrication procedures and objectives reflect the ideal for the plant, its personnel and the bottom line.



Want to learn more about this course?
Download our brochure or visit **Noria.com/train**

Training Options

- Public
- Onsite

Course Topics

- Lubricant Health Monitoring
- Oil Analysis Fundamentals
- Oil Sampling – The Very Best Practices
- Fluid Properties Analysis
- Contamination Control and Proactive Maintenance
- Fault Detection and Wear Particle Analysis
- Instrument-free Onsite Tests
- Interactive Case Studies Workshop
- Test Results and Oil Analysis Report Interpretation

Practice In Class

Oil Analysis II includes practice oil analysis reports and individual and group participation in problem-solving exercises.

Get Certified

Oil Analysis II prepares students for ICML Level II MLA certification. Certification testing is available the Friday following the training. Register for your exam at **LubeCouncil.org**.

“ This is an outstanding course in understanding the desired state and use of oil and how to determine the best practices. I wish I would have taken it a year and a half ago. ”

– Derik Williams, Engineer, US Army CERDEC



Take This Course If You Are Responsible For:

- Resource management
- Equipment maintenance
- Systems operation/management
- Lubricant health and cleanliness targets
- Mechanical inspections
- Diagnostics and equipment troubleshooting
- Efficiency initiatives/design
- Quality-control implementation
- Operations planning
- Sensor selections/installation
- Lubrication program effectiveness
- Choosing test slates

Don't Forget Your Study Packet

The Level II MLA Certification Study Packet is designed to prepare students for ICML Level II MLA certification. The packet includes:

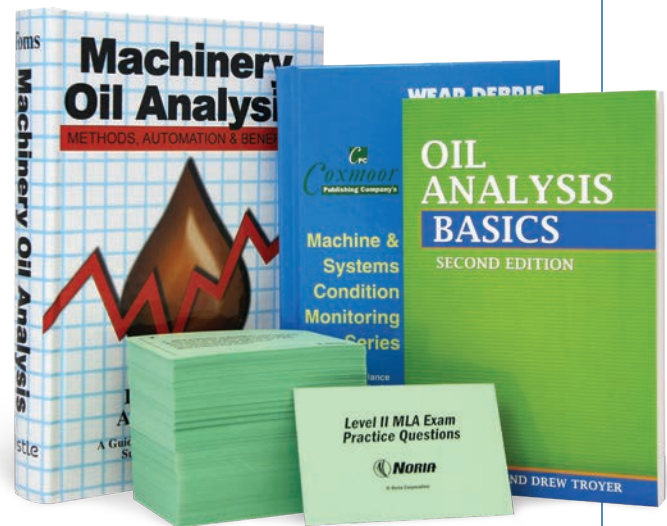
Level II MLA Flash Cards - 440+ flash cards designed in preparation for the ICML Level II MLA certification exam.

Oil Analysis Basics - With 90+ illustrations, figures and look-up tables, this book presents the fundamentals of oil analysis for machinery condition monitoring in an easy-to-understand format. The book covers everything from how to take a proper oil sample to how to select a test slate for your applications.

Wear Debris Analysis - With 70+ illustrations, figures and tables, this book offers a practical look at wear debris and wear particle analysis in many forms.

Machinery Oil Analysis - This unique, in-depth analysis presents the entire practice of oil analysis as a condition monitoring tool for machines. The book also describes the what, when, where and how-to for machinery lubrication concepts, machinery failure and maintenance concepts, machinery failure modes, oil sampling and testing, plus statistical analysis and data interpretation.

Add the Study Packet to your course registration, or visit us at store.noria.com.





Master Advanced Analysis

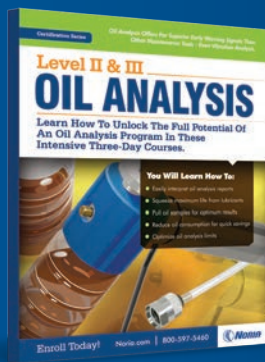
OIL ANALYSIS III

Take the Next Steps in Transforming Your Oil Analysis

Designed as a more in-depth look into this specialized field, Oil Analysis III (OA III) gives you the tools and education to develop, implement and sustain a first-rate, comprehensive oil analysis program. Merging the machine lubrication and oil analysis focuses of course prerequisites, OA III trains you to diagnose problems like contamination, corrosion and additive depletion – even with limited test results or difficult machines.

Through this course, you will compare detection and cost benefits/drawbacks for any type of lab test to select the most precise option for your needs. You will be able to gather extensive data on a lubrication program, make informed decisions and maintain the program as a major company asset. You will also gain strategies for applying your data and analysis to build strong proposals for improvement projects and calculate accurate return on investment predictions.

By completing OA III, you will attain the highest level of expertise and join the elite ranks of others who create or assess new oil analysis programs within any industrial workplace.



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MAINTAINABILITY CENTER

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CREDIT
APPROVED

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or visit **Noria.com/train**

Training Options

- Public
- Onsite

Course Topics

- Understanding and Analyzing Machine Wear
- Integrating Oil Analysis with Vibration Analysis
- Fluid Properties Analysis
- Additive Depletion
- Contamination Analysis
- Grease Analysis
- Onsite Oil Analysis Options
- Designing an Oil Analysis Program
- Cost/Benefit Analysis

Practice In Class

Oil Analysis III includes practice oil analysis reports and individual and group participation in problem-solving exercises.

Get Certified

Oil Analysis III prepares students for ICML Level III MLA certification. Certification testing is available the Friday following the training. Register for your exam at **LubeCouncil.org**.

“I was amazed of the information that’s out there, and the techniques or **tests that can be taken for oil analyses.**”

-Reuben Clowdis, Lubrication Tech *Huber Engineered Woods, LLC*



Take This Course If You Are Responsible For:

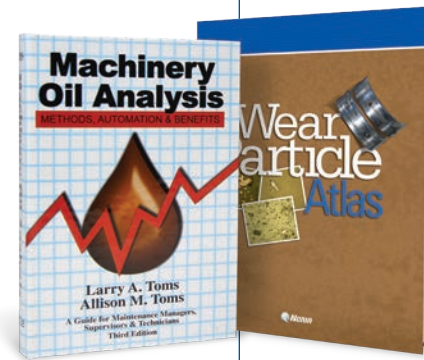
- Systems operation/management
- Operations
- Equipment maintenance
- Testing samples and interpreting results
- Mechanical inspections
- Diagnostics
- Troubleshooting equipment design/process/material
- Efficient utilization design
- Procedure/quality-control implementation
- Cost/benefit analysis
- Quality-control analysis

Suggested Reading

Machinery Oil Analysis Methods, Automation & Benefits - This book uniquely presents the entire practice of oil analysis as a condition monitoring tool for machines.

Wear Particle Atlas - A guide to wear particle analysis, this guide contains photographs of typical wear particles found in used lubricating oil, illustrative case histories and operational procedures for wear debris analysis.

You can purchase these books online at **Store.Noria.com**.



Avoid Costly Mistakes And Reap The Benefits Of Good Oil Sampling

A small misstep when installing sampling ports, performing oil sampling or choosing lubricants can translate into expensive problems. For most facilities, these mistakes are only apparent after problems arise or through expert analysis. For example, a large steel mill installed 1,200 new sampling ports and, after gathering strange data on multiple occasions, learned they had been placed in the wrong locations.

Another Noria customer had been conducting oil analysis for years but was still unable to detect bearing failures that shut down

production. The company discovered that the tests being conducted could only identify degraded or wrong lubricants – not impending failure.

Successful oil sampling requires know-how.

Improper sampling port placement, running the wrong tests, poorly collected samples and more can compromise accuracy or even eliminate valuable information. Noria can lend you the expertise to avoid these costly mistakes so you can reap the full benefits of good oil sampling.



Services & Solutions

Lubrication Program Development

Noria's Lubrication Development Program (LPD) is a comprehensive lubrication overhaul of your facility. Starting with a 540-point assessment of your existing lubrication practices and materials, we use our findings to detail your "roadmap" to lubrication excellence. After developing a custom lubrication program, Noria experts then deliver detailed instructions for maintenance routes, hardware modifications, lubricant types and procedures. We also make recommendations for redesigning your lubricant storage area to preserve lubricant quality and maximize space.



Prevent Mechanical Failure,
Train Personnel And
Maximize Efficiency
In One Program.



For facilities needing to tap into Noria's network of skilled lubrication and oil analysis professionals, Workforce Solutions is the lubrication labor outsourcing division of Noria Corporation. We place skilled personnel at your facilities on a full-time or short-term basis to manage your lubrication program and/or execute aspects of machinery lubrication and oil analysis for more details visit:

Noria.com/serve/workforce-solutions



Education & Training

To achieve plant-wide lubrication excellence, Noria can train all personnel – from those with no technical background to those with specialized roles. Noria offers five nationally recognized courses on machinery lubrication and oil analysis, with four of these containing certification preparation. For more details visit:

Noria.com/train



Technical Consulting

Lubrication-enabled reliability is Noria's core mission; we supply answers and advice for any reliability issue about lubrication. Over the years, Noria technical consultants have provided expert witnessing and forensic investigations, laboratory studies, benchmarking and specialized troubleshooting on a variety of machine parts and systems.

For More Information,
Call 800-597-5460

Failure Investigation

Machine and lubricant failures-those are words no one likes, but there is a solution. Noria's network of expert investigators covers a multitude of industries and applications. Our technical consulting team has decades of experience in root cause analysis, tribological studies, lubricant analysis, debris/wear particle identification, metallography and more. Our failure investigators will provide the answers and solutions to quickly and efficiently identify the cause to ensure a complete and permanent fix.





ONLINE LEARNING

We realize schedules get hectic. That's why Noria offers two of its most popular courses, Machinery Lubrication I and Machinery Lubrication II, online. With our online learning, you receive:

- **Flexible Certification Preparation:** Gain the right understanding and steps for excellent condition-based maintenance, prepare for a certification test and do it all on your own time – whenever it is best for you.
- **Onboarding:** Use this training for onboarding new employees or to get your team up to speed quickly, efficiently and in alignment with your company culture, skills requirements and performance expectations.
- **Self-paced:** Log in anywhere, anytime on your customized schedule for one year. Play, pause, rewind and replay as often as needed. Learn while maintaining plant and operational schedules. ADA-compliant closed captioning is available.
- **Pre-/Post-assessment Quizzes:** Track your knowledge and understanding through each section with quizzes designed to show your progress and areas that may require further attention.

For more information, visit Noria.com/train or call 918-392-5063.

Available Courses

- Machinery Lubrication I
- Machinery Lubrication II

Key Features

- 125-question pre-and post-assessments
- Knowledge checks for each section to evaluate your understanding. Know what to work on before moving forward.
- Step-by-step interactive demonstrations of common lubricant analysis tests
- Meets the training requirements for the ICML MLA and MLT Level I certifications
- SCORM compliant, tablet ready
- Includes closed captioning

Why Train Online?

- Affordable:** Reduce training costs and improve productivity.
- Flexible:** Anytime, anywhere. Online training enables your team to learn at their own pace.
- Easy To Use:** Noria's simple click-and-watch training method is easy for anyone to use.
- Course Manual:** Downloadable PDF manual for future reference.



ENTERPRISE ONLINE LEARNING

Private Onsite Training

Bring online machinery lubrication and reliable skills training to your entire company through Enterprise Online Learning. Our team works closely with you to create a learning plan that meets your short- and long-term goals.

Noria can host training content on your learning management system (LMS) or provide a private, easy-to-use learning portal for your team. Noria's online learning solutions provide the following features:

- Monitor learning progress
- Train on your team's schedule
- Integrate machinery lubrication reliability to your training portfolio
- Verify skills
- Prepare for industry certification
- Reduce travel expenses
- Onboard new employees faster
- Improve efficiencies and reduce costly errors
- Train your entire workforce

All modules are compliant with all training industry standards, and we can successfully implement the following reporting formats: AICC, SCORM 1.2 and xAPI (Tin Can). If you don't have an LMS, ask us about a no-hassle, branded solution for your organization.

BRING NORIA TRAINING TO YOU

Available Courses

- Industrial Lubrication Fundamentals
- Machinery Lubrication I & II
- Oil Analysis II & III

Private Onsite Training

Looking for a personalized training plan for your facility? Noria offers onsite training to meet your company's schedules, goals and budget needs. Additionally, Noria can customize any of its courses based on your specific needs.

By providing onsite training, Noria helps employers avoid travel time/costs. This option also allows for personalized training with examples from onsite equipment. Many of our customers have utilized this option to rotate groups of employees through consecutive one-day trainings and to bring personnel from multiple plants to one onsite training.

Whether you have 15 people or 500, Noria is committed to providing options that will have the smallest impact on your production.

For more information, visit [Noria.com/train](https://noria.com/train) or call 918-392-5063.



RELIABILITY SKILLS SERIES

Reliability Skills Series Online

Need fundamental and best-practice training for everyone in your facility? Noria's Reliability Skills Series provides online lessons to help your facility rise to the top.

How to Use a Grease Gun delves into best practices for using a grease gun. Although it's one of the most widely used tools for machinery lubrication, few are properly trained to use it. When used or loaded improperly, the grease gun becomes a safety risk to both the lubrication technician and machine.

How to Grease a Motor Bearing gives plant personnel an overview of the best practices for lubricating electric motor bearings. Use this evergreen course to train operators, lubrication technicians, mechanics, electricians and maintenance personnel for years to come.

Introduction to Lubrication Fundamentals presents a solid foundation of the basics of industrial lubricants and tribology, including oil and grease properties, additives, synthetics, mineral oils, lubricant oxidation and other important topics.

Best Practices for Oil Sampling shows you how to design and implement a world-class oil sampling program that will deliver better results and help you focus on improving equipment reliability.

Best Practices for Lubricant Storage and Handling provides procedures to apply right away. From delivery to dispensing to filling the machine, you will learn the best practices for new oil storage and handling.

Lubrication Basics for Machinery Operators offers basic lubrication training, including how to recognize the early signs of lubrication-related problems, grease gun basics and safety, checking oil levels, cleaning and inspecting machinery, oil leak inspections and more.

Add this online series to your course registration or visit store.noria.com.



REGISTRATION FORM

HOW TO REGISTER

ONLINE

Noria.com/training

FAX (3-4 days processing)

918-746-0925

PHONE

800-597-5460
or
918-749-1400

MAIL (3-4 weeks processing)

Noria Training
1328 E. 43rd Court
Tulsa, OK 74105

PLEASE PRINT CLEARLY

Last Name	First Name	Middle Initial
Job Title	Organization/Business Name	
Mailing Address		
City	State/Province	Postal Code/ZIP
Country		
Phone	Fax	
Email (required)		

Note: The mailing address is where all mailed correspondence about this registration will be sent. The email address is how Noria communicates important program details to you.

PLEASE REGISTER ME FOR THE FOLLOWING COURSE(S):

COURSE DATE

LOCATION

☐ Machinery Lubrication I			\$1,295
☐ Optional: Study Packet - Level I MLT / Level I MLA Price includes shipping to a U.S. address			\$295
☐ Machinery Lubrication II			\$1,295
☐ Oil Analysis II			\$1,295
☐ Optional: Study Packet - Level II MLA Price includes shipping to a U.S. address			\$310
☐ Oil Analysis III			\$1,295

Note: Payment is due before the course. Total Amount Due \$ _____

- ☐ Check is enclosed. (Allow 3-4 weeks for processing.) Please make checks payable to Noria Corporation in U.S. dollars drawn on a U.S. bank.
- ☐ I authorize Noria Corporation to charge \$ _____ to my: _____
- ☐ VISA ☐ MasterCard ☐ AMEX ☐ Discover

Card Number _____ Expiration Date _____

Signature _____

Print name as it appears on credit card bill _____

☐ Our purchase order is attached. P.O. Number _____

CANCELLATIONS AND SUBSTITUTIONS If you cannot attend the course, a colleague can attend in your place. Registrations canceled at any time prior to 10 days before the course are not subject to fees. Cancellations after 10 days are subject to a \$75 service charge. Or, your registration fee can be transferred to another program of your choice to be taken within 12 months. For more information, visit Noria.com/train/registration.

CERTIFICATION EXAMS Certification testing is offered by the International Council for Machinery Lubrication the morning following a training course at the same hotel. Please contact ICML to register for the certification exam or register online at LubeCouncil.org.

SATISFACTION GUARANTEE Noria proudly stands behind our training courses. If you're not satisfied with the information and insights you gain at these events, we'll refund 100% of your paid registration fee.

Register online at Noria.com/training or call 800-597-5460





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TRAINING THAT WORKS

Find the training that works for you. Noria's machinery lubrication and oil analysis training courses offer something for all levels of expertise, from beginners to experienced technicians and engineers. Enroll today to learn precision lubrication skills for maximizing machine reliability. **For dates and locations, check out [Noria.com/train](https://noria.com/train).**

